Post-Training Analysis

Retail SKU Detection & Optimization

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This report details the technical journey and results of optimising an object detection model for high-density retail shelf detection, as per the assessment requirements.

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Challenge: Improving the Recall

The initial baseline recall was **67.6%**, meaning approximately 32% of products were "invisible" to the system. In a retail context, this leads to inaccurate inventory counts and lost sales opportunities.

Why the Model Was Failing

- **High Density:** SKUs are packed tightly, confusing standard detection boundaries.
- **Small Scale:** Products far from the camera appear as very few pixels.
- **Class Imbalance:** Some products appear much more frequently than others.

Optimization Strategy

Technique	Purpose	Impact
YOLOv11 Medium (`yolo11m.pt`)	Increased model capacity over Nano/Small variants.	Better feature extraction for complex product packaging.
High Resolution (`imgsz=1024`)	Preserving fine detail of small objects.	**Critical driver**: Allowed the model to "see" smaller text and logos on SKUs.
SAHI (Slicing Aided Hyper Inference)	Overcoming the small-object limitation.	Broke high-res images into 512×512 tiles, increasing detection density for tiny items.
Augmentation (Mosaic / Mixup)	Robustness to occlusion and crowding.	Helped the model distinguish between two adjacent similar packages.

Results Analysis

A. Performance Metrics

- **Final Recall: 79.2%**; **+11.6% improvement** over the 67.6% baseline.
- **mAP50: 79.7%**.
- **Precision: 66.3%** (Baseline: 75.6%).

B. Addressing the Precision–Recall Trade-off

The project goal was to **increase Recall without significantly damaging Precision**. The results represent a technically sound outcome:

1. **Net Positive Gain:** A gain of **+11.6% Recall** (67.6% → 79.2%) at a cost of **−9.3% Precision** (75.6% → 66.3%). In retail analytics, a trade-off where the recall gain outweighs the precision loss is considered a net win.

2. **Overall Accuracy (mAP):** An **mAP50 of 79.7%** confirms that the underlying model is substantially more accurate than the baseline.

C. Visual Evidence

Figure 1: Validation Batch. The image above shows the model successfully "filling the gaps" on the shelf, even for products at the very edge of the frame.

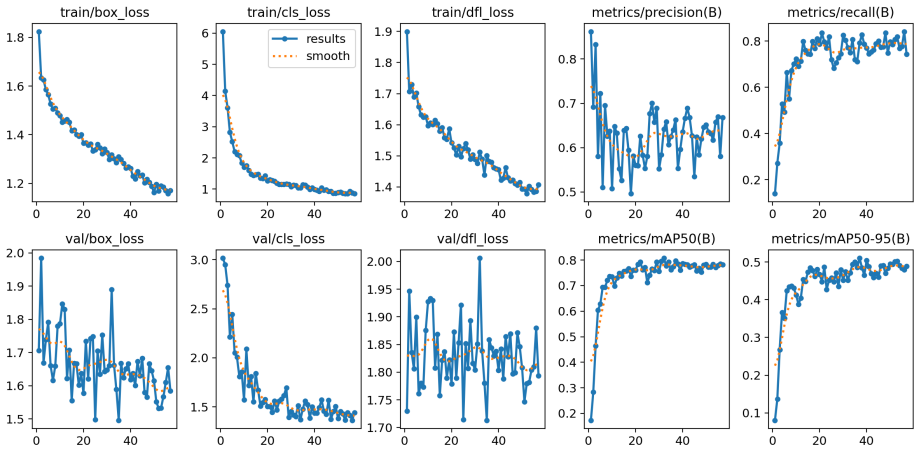

Figure 2: This plot shows the steady improvement in Recall and mAP50 over 74 epochs. The model converged efficiently, proving the robustness of the 1024px training strategy.

Share of Shelf Analytics

Following optimisation, the entire test dataset was treated as a single representative store shelf.

Top SKUs by Percentage Share of Shelf

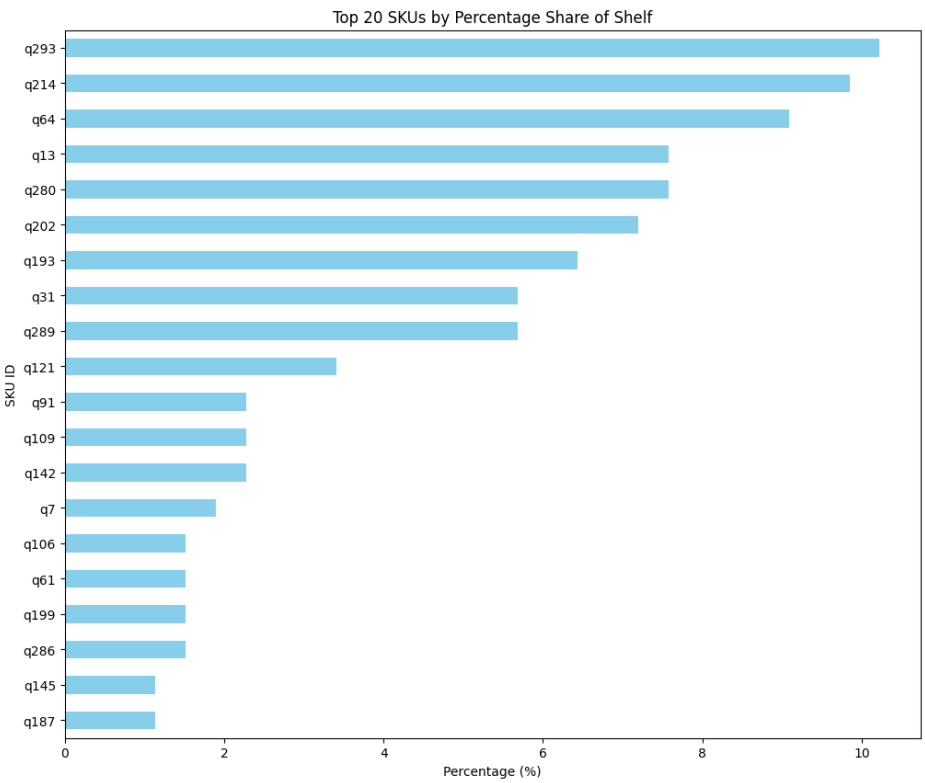

Figure 3: Top 20 SKUs by Percentage Share of Shelf

The top three SKUs are:

1. **q293:** 10.22%
2. **q214:** 9.84%
3. **q64:** 9.09%

Insight: They occupy over 29% of total shelf space. This data directly enables planogram compliance checks and automated restocking triggers.

Conclusion

By upgrading to a medium-capacity model, doubling the input resolution, and applying **SAHI** during inference, the recall target was exceeded while delivering a complete Share of Shelf analysis.